



Lecture 1 — Before class

Introduction to Quantitative Biology | Read before Thursday

This course is **flipped**: before each lecture you receive a short sheet like this one, and in class we use it rather than repeat it. For the first lecture you have three things to do.

1. **Review ordinary differential equations** (Section 1). There are short exercises, with solutions at the end, so you can check yourself.
2. **Read Chapter 1 of Uri Alon's book** [1], as far as you can (Section 2 tells you which parts matter most).
3. **Write down your answers to the questions in Section 4**, and bring them. We start the lecture from them.

You do not need to understand everything. Coming with a precise question ("I do not see why the response time does not depend on β ") is as useful as coming with an answer.

All PDFs are on the GitHub page of the course: <https://github.com/qbiomaster/qBioMaster-introduction>.

What we will do on Thursday

- What quantitative biology is, and how this course works.
- Some numbers of a cell: how many proteins, how long they live, how long a cell cycle lasts.
- The simplest model of a gene: a protein that is produced and removed. We build the equation together, solve it, and find its response time.
- The same gene described in different ways: a differential equation, random events in a single cell, a distribution over many cells. Each answers a different question.
- How proteins are shared between the two daughter cells when a cell divides, and why this makes genetically identical cells different. On Friday, in InLab, you will simulate it.

1. Reviewing ordinary differential equations

What an ODE is

Many quantities in biology change in time: the number of proteins in a cell, the number of bacteria in a flask, the concentration of a drug in the blood. Call this quantity $x(t)$. Often we do not know $x(t)$ directly, but we know *how fast it changes* given its current value. We write this as

$$\frac{dx}{dt} = f(x).$$

On the left is the rate of change of x (the derivative with respect to time). On the right is a function f that depends on the current value of x . An equation of this kind is an **ordinary differential equation** (ODE). "Ordinary" means that there is only one variable we differentiate with respect to, here the time t .

Solving the ODE means finding a function $x(t)$ whose derivative equals $f(x(t))$ at every time. To pick one solution we also need to know where we start: the **initial condition** $x(0) = x_0$.

Checking a solution is always easy, even when finding it is hard: compute the derivative of your candidate $x(t)$, and verify that it equals $f(x(t))$ and that $x(0) = x_0$.

Two facts you will need

- The derivative of e^{bt} is $b e^{bt}$.
- $\ln$ is the natural logarithm: $\ln(e^a) = a$, and $\ln 2 \approx 0.693$. Alon writes it $\log$.

Example 1: constant change

If x grows at a constant rate c ,

$$\frac{dx}{dt} = c, \quad x(0) = x_0,$$

the solution is a straight line, $x(t) = x_0 + ct$. Check it: the derivative of $x_0 + ct$ is c , and at $t = 0$ it gives x_0 .

Example 2: exponential decay

Suppose each molecule has the same probability per unit time k of disappearing. The more molecules there are, the faster their number decreases:

$$\frac{dx}{dt} = -kx, \quad x(0) = x_0.$$

Method 1 — guess. Try $x(t) = Ce^{bt}$. Its derivative is $bCe^{bt} = bx$. For this to be $-kx$ we need $b = -k$. The initial condition gives $C = x_0$. So

$$x(t) = x_0 e^{-kt}.$$

Method 2 — separation of variables. Move everything that contains x to the left, and everything that contains t to the right:

$$\frac{dx}{x} = -k dt.$$

Integrate both sides, the left from x_0 to $x(t)$ and the right from 0 to t :

$$\ln x(t) - \ln x_0 = -kt \implies x(t) = x_0 e^{-kt}.$$

The two methods give the same answer.

Half-life. The time $t_{1/2}$ after which half of the molecules are left is found from $x(t_{1/2}) = x_0/2$:

$$e^{-kt_{1/2}} = \frac{1}{2} \implies t_{1/2} = \frac{\ln 2}{k}.$$

It does not depend on x_0 : starting from 10 or from 10 000 molecules, half of them are gone after the same time.

Reading an ODE without solving it

Very often we cannot solve an ODE, but we can still say a lot by looking at the *sign* of dx/dt :

- where $f(x) > 0$, x increases;
- where $f(x) < 0$, x decreases;
- where $f(x) = 0$, x does not change. These values are the **steady states** (or fixed points).

Drawing arrows on the x axis (up where $f > 0$, down where $f < 0$) shows where the system goes from any starting point.

Exercise 1: Check yourself

Solutions are at the end of this sheet.

- (a) What are the units of k in $dx/dt = -kx$, if x is a number of molecules and t is in minutes?
- (b) A protein is degraded with $k = 0.1 \text{ min}^{-1}$. What is its half-life? Starting from $x_0 = 100$ molecules, how many are left after 20 minutes?
- (c) Check, by computing the derivative, whether each function is a solution of $dx/dt = -2x$ with $x(0) = 3$: (i) $x(t) = 3e^{-2t}$; (ii) $x(t) = e^{-2t} + 2$.
- (d) Consider $dx/dt = x(1-x)$ for $x \geq 0$ (a population that grows but is limited by resources, with x measured in units of the maximal population). Without solving it: find the steady states; say whether x increases or decreases for $0 < x < 1$ and for $x > 1$; sketch $x(t)$ starting from $x_0 = 0.1$ and from $x_0 = 1.5$.

Exercise 2: A question for Thursday

In Example 2 we separated the variables of $dx/dt = -kx$. Try the same steps on the *general* equation

$$\frac{dx}{dt} = f(x), \quad x(0) = x_0.$$

What do you obtain? Is it a formula for $x(t)$? What could stop you from going further for a given f ? There is no solution at the end: we discuss it in class.

2. Reading Alon, Chapter 1

Chapter 1 of *An Introduction to Systems Biology* [1] introduces transcription networks and, in its Section 1.4, the model we start from on Thursday. Read as much as you can, in this order of priority:

1. **Section 1.4 and Section 1.4.1** (pp. 13–15) — read carefully, pen in hand. Try to redo the calculations yourself.
2. **Sections 1.1 to 1.3.2** (pp. 3–8) — read quickly. Look at the numbers in Table 1.1 and Table 1.2: we use them in class.
3. Sections 1.3.3 to 1.3.5 (input functions, Hill functions) — optional for now. We come back to them in a later lecture.

Notation

We use Alon's notation. Y is the concentration of a protein, β its production rate, and α its removal rate (degradation plus dilution by cell growth):

$$\frac{dY}{dt} = \beta - \alpha Y.$$

Section 1 used x and k for a general equation; in Alon's model, Y plays the role of x and α plays the role of k . Other books write the same model with different letters.

3. Optional reading

Not needed to follow Thursday's lecture, but useful if you have time:

- Phillips *et al.*, *Physical Biology of the Cell*, Chapter 2, pp. 43–48 [2]: how molecules are shared between daughter cells at division. It is also the starting point of Friday’s InLab session.
- Wilkinson (2009) [3]: only the box on modelling, which compares different descriptions of the same system.

4. Questions to bring to class

Exercise 3: Write down your answers

These questions use Section 1 and Alon, Section 1.4. Short answers are enough.

1. In $dY/dt = \beta - \alpha Y$, Y is a concentration. What are the units of β and of α ?
2. Two genes have the same removal rate α , but the second is produced ten times faster than the first ($\beta_2 = 10\beta_1$). Both start from $Y = 0$. Which one has the higher steady state? Which one reaches half of its own steady state first?
3. A stable protein (not degraded) in an *E. coli* cell that divides every 30 minutes: what removes it, what is α , and what is its response time (Alon, Section 1.4.1)?
4. Alon writes that some proteins are produced and degraded quickly, in a “futile cycle”. What does the cell gain from this, and what does it pay?
5. According to Table 1.1, one protein per *E. coli* cell corresponds to a concentration of about 1 nM. A transcription factor is present at 5 nM: how many copies is that per cell? Do you think a smooth curve $Y(t)$ describes well what happens in *one* such cell? (There is no single right answer yet.)
6. Your answer to “A question for Thursday” in Section 1.

Solutions to “Check yourself”

- (a) dx/dt is in molecules per minute and x in molecules, so k is in min^{-1} : a probability per unit time.
- (b) $t_{1/2} = \ln 2/0.1 \approx 6.9$ min. After 20 min, $x = 100 e^{-0.1 \times 20} = 100 e^{-2} \approx 13.5$, so about 13 or 14 molecules.
- (c) (i) Yes: the derivative is $-6e^{-2t} = -2 \times 3e^{-2t} = -2x$, and $x(0) = 3$. (ii) No: the derivative is $-2e^{-2t}$, while $-2x = -2e^{-2t} - 4$. (It does satisfy $x(0) = 3$, which shows that the initial condition alone is not enough.)
- (d) Steady states where $x(1-x) = 0$: $x = 0$ and $x = 1$. For $0 < x < 1$ both factors are positive, so x increases; for $x > 1$ the factor $(1-x)$ is negative, so x decreases. Starting from 0.1, x rises towards 1; starting from 1.5, it falls towards 1. Neither curve crosses $x = 1$.

References

- [1] Uri Alon. *An Introduction to Systems Biology: Design Principles of Biological Circuits*. 2nd. CRC Press, 2019.
- [2] Rob Phillips et al. *Physical Biology of the Cell*. 2nd. Garland Science, 2013.
- [3] Darren J. Wilkinson. “Stochastic modelling for quantitative description of heterogeneous biological systems”. In: *Nature Reviews Genetics* 10.2 (2009), pp. 122–133. DOI: [10.1038/nrg2509](https://doi.org/10.1038/nrg2509).